

Spanish American biochemist Severo Ochoa was awarded the Nobel Prize for discovering a bacterial enzyme that synthesizes ribonucleic acid (RNA)—a molecule that is essential to all forms of life. His findings were partly flawed, but they were valuable, as they revealed more of the process by which genetic information is encoded in cells.

MILESTONES

CAJAL'S INFLUENCE

His interest in biology is inspired by Santiago Ramón y Cajal's studies of the nervous system.

ANALYZES ENZYMES

Finds a bacterial enzyme that appears to synthesize RNA in 1955, while researching phosphates.

RECEIVES NOBEL PRIZE

Wins joint Prize in 1959 for work on RNA; is the first Hispanic American to receive the honor.

SUFFERS SETBACK

Enzyme shown in 1955 not to synthesize RNA, but to degrade it, although in a test tube this is reversed.

Born in Madrid, Severo Ochoa held posts at European and US universities before becoming professor of biochemistry at New York University, which was renowned for its work in enzymology. In 1954, the year after Francis Crick and James Watson demonstrated the helix structure of DNA (see pp.264–269), Ochoa became interested in deciphering the genetic code and decided to investigate the mechanism for synthesizing nucleic acids.

In 1955, after a complex process of elimination in his laboratory, Ochoa chose the *Azotobacter vinelandii* bacterium to isolate an enzyme that he called polynucleotide phosphorylase. In test-tube conditions, the enzyme was found to synthesize ribonucleic acid, or RNA—a molecule that, like DNA, plays an important role in the transfer of genetic information.

Discovery debated

While Ochoa was conducting his experiments, the US biochemist Arthur Kornberg achieved a similar result with a different bacterium. The enzyme that Kornberg isolated appeared to explain the mechanism for synthesizing DNA. The two men shared the 1959 Nobel Prize in Physiology or Medicine for their discoveries, but soon afterward, observations revealed that the roles assigned to the two enzymes could not be confirmed. Nevertheless, Ochoa's enzyme was helpful in his later work on the genetic code.

Ochoa and his team were delighted to hear of the Nobel Prize nomination for their work, which showed that RNA can be synthesized artificially.

“Polynucleotide phosphorylase may be considered ... the **Rosetta Stone of the genetic code.”**

Severo Ochoa, 1980

